

Worked examples for Lecture 14: Seismic sources

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These examples accompany Lecture 14. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: The stress glut of a fault in an isotropic body, and the moment tensor. An idealised fault lies in the (x_1, x_2) -plane of an isotropic body with Lamé parameters λ and μ , and the displacement discontinuity vector $\mathbf{w}$ is tangential to the fault, so that $w_3 = 0$.

- (a) Starting from the stress glut $\bar{S}_{ij} = A_{ijk}w_k\delta(x_3)$ obtained in the lecture, show that

$$\bar{S}_{ij} = \mu (w_i\delta_{j3} + w_j\delta_{i3}) \delta(x_3),$$

and comment on the absence of λ .

- (b) The **moment tensor** of a source is defined as the volume integral of the stress glut,

$$M_{ij}(t) = \int_M \bar{S}_{ij}(\mathbf{x}, t) d^3\mathbf{x}.$$

Suppose that for $t > 0$ the slip $\mathbf{w} = D \hat{\mathbf{e}}_1$ is uniform over a patch Σ of the fault having area $\mathcal{A}$, that μ is constant on Σ , and that Σ is small compared with the wavelengths of interest. Explain why the stress glut can then be replaced by $M_{ij}\delta(\mathbf{x})$, determine M_{ij} , and find its eigenvalues and eigenvectors. Express $\mathbf{M}$ in terms of its eigenvectors.

Solution. (a) For an isotropic material the elastic tensor takes the form

$$A_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \quad (1)$$

and so, setting $l = 3$ and contracting with w_k ,

$$A_{ijk}w_k = \lambda\delta_{ij}\delta_{k3}w_k + \mu(\delta_{ik}\delta_{j3} + \delta_{i3}\delta_{jk})w_k = \lambda w_3\delta_{ij} + \mu(w_i\delta_{j3} + w_j\delta_{i3}). \quad (2)$$

The first term is proportional to w_3 , which vanishes because the displacement discontinuity is tangential to the fault, and hence

$$\bar{S}_{ij} = \mu(w_i\delta_{j3} + w_j\delta_{i3})\delta(x_3). \quad (3)$$

Written out, the only non-zero components are

$$\bar{S}_{13} = \bar{S}_{31} = \mu w_1\delta(x_3), \quad \bar{S}_{23} = \bar{S}_{32} = \mu w_2\delta(x_3), \quad (4)$$

so that the stress glut is symmetric, as it must be within the linearised theory, and also traceless, $\bar{S}_{ii} = 2\mu w_3 \delta(x_3) = 0$. The same calculation for a general fault surface with unit normal $\hat{\mathbf{n}}$ gives $\bar{S}_{ij} = \mu(w_i \hat{n}_j + w_j \hat{n}_i) \delta_\Sigma$ whenever $\mathbf{w} \cdot \hat{\mathbf{n}} = 0$. The Lamé parameter λ enters only through the term $\lambda w_3 \delta_{ij}$, which is to say only if the fault opens or the two sides interpenetrate. For a shear dislocation the source is therefore completely insensitive to λ ; the radiated waves depend on λ only through their propagation, not through their generation.

(b) The stress glut enters the linearised equations of motion through $-\partial \bar{S}_{ij} / \partial x_j$, and it acts on the displacement field only through integrals against smooth functions. For any smooth ϕ , using the defining property of $\delta(x_3)$ to reduce the volume integral to one over the fault,

$$\int_M \bar{S}_{ij} \phi \, d^3 \mathbf{x} = \int_\Sigma \mu (w_i \delta_{j3} + w_j \delta_{i3}) \phi \, dS \approx \phi(\mathbf{0}) \int_\Sigma \mu (w_i \delta_{j3} + w_j \delta_{i3}) \, dS = \phi(\mathbf{0}) M_{ij}, \quad (5)$$

provided that ϕ varies little over Σ , which is the case for wavefields whose wavelengths are large compared with the patch. Since $\int_M M_{ij} \delta(\mathbf{x}) \phi \, d^3 \mathbf{x} = M_{ij} \phi(\mathbf{0})$, the replacement $\bar{S}_{ij} \rightarrow M_{ij} \delta(\mathbf{x})$ preserves precisely this zeroth moment of the glut, and this is the **point-source approximation**. Evaluating the integral for uniform slip $\mathbf{w} = D \hat{\mathbf{e}}_1$ and constant μ gives, for $t > 0$,

$$M_{ij} = \mu \mathcal{A} D (\delta_{i1} \delta_{j3} + \delta_{i3} \delta_{j1}), \quad \text{i.e.} \quad \mathbf{M} = M_0 \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad M_0 = \mu \mathcal{A} D, \quad (6)$$

where M_0 is the **scalar seismic moment**, with units of N m. A moment tensor of this form is called a **double couple**, for reasons made clear in the next example.

To find the eigenvalues we expand $\det(\mathbf{M} - \sigma \mathbf{I})$ along its second row, which contains the single non-zero entry $-\sigma$, to obtain

$$\det(\mathbf{M} - \sigma \mathbf{I}) = -\sigma (\sigma^2 - M_0^2), \quad (7)$$

so the eigenvalues are $M_0, 0$ and $-M_0$. The corresponding unit eigenvectors are found by inspection, since $\mathbf{M}$ simply exchanges the first and third components of a vector (scaled by M_0) and annihilates the second,

$$\hat{\mathbf{t}} = \frac{1}{\sqrt{2}} (\hat{\mathbf{e}}_1 + \hat{\mathbf{e}}_3), \quad \hat{\mathbf{b}} = \hat{\mathbf{e}}_2, \quad \hat{\mathbf{p}} = \frac{1}{\sqrt{2}} (\hat{\mathbf{e}}_1 - \hat{\mathbf{e}}_3), \quad (8)$$

with $\mathbf{M} \hat{\mathbf{t}} = M_0 \hat{\mathbf{t}}$, $\mathbf{M} \hat{\mathbf{b}} = \mathbf{0}$ and $\mathbf{M} \hat{\mathbf{p}} = -M_0 \hat{\mathbf{p}}$. The vector $\hat{\mathbf{t}}$ lies at 45° to the fault plane in the direction of slip and is known as the **tension** or T-axis; $\hat{\mathbf{p}}$, also at 45° to the fault, is the **pressure** or P-axis; and $\hat{\mathbf{b}}$, which lies in the fault plane perpendicular to the slip, is the **null** or B-axis. Because $\mathbf{M}$ is symmetric, it has the spectral decomposition

$$\mathbf{M} = M_0 (\hat{\mathbf{t}} \hat{\mathbf{t}}^T - \hat{\mathbf{p}} \hat{\mathbf{p}}^T), \quad (9)$$

which can be checked directly since $\hat{\mathbf{t}} \hat{\mathbf{t}}^T - \hat{\mathbf{p}} \hat{\mathbf{p}}^T = \hat{\mathbf{e}}_1 \hat{\mathbf{e}}_3^T + \hat{\mathbf{e}}_3 \hat{\mathbf{e}}_1^T$. Eq. (9) says that a double couple is equivalent to a pair of equal and opposite linear dipoles, a tensile one along $\hat{\mathbf{t}}$ and a compressive one along $\hat{\mathbf{p}}$, at 45° to the fault.

Two features of the result are worth noting. First, the trace of $\mathbf{M}$, which is the sum of its eigenvalues, vanishes; a double couple involves no change in volume. Second, $\mathbf{M}$ in eq. (6) is unchanged if $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_3$ are interchanged, i.e. if the slip direction and the fault normal swap roles. Slip $D\hat{\mathbf{e}}_3$ on a fault with normal $\hat{\mathbf{e}}_1$ produces an identical moment tensor. This second plane is known as the **auxiliary plane**, and the moment tensor alone cannot distinguish it from the true fault plane; both planes contain the null axis and bisect the P- and T-axes.

Example 2: The equivalent body force for a point source. A point source in the interior of a body has stress glut $\bar{S}_{ij}(\mathbf{x}, t) = M_{ij}(t)\delta(\mathbf{x})$, where the moment tensor M_{ij} is symmetric.

- (a) Show that the linearised equations of motion and boundary conditions are those of the same body subject to a body force $f_i = -M_{ij}\partial\delta(\mathbf{x})/\partial x_j$ and no applied surface traction.
- (b) Show that this force exerts no net force and no net torque on the body,

$$\int_M f_i d^3\mathbf{x} = 0, \quad \int_M \epsilon_{ijk} x_j f_k d^3\mathbf{x} = 0,$$

and identify which property of M_{ij} is responsible for each.

- (c) For the double couple of Example 1, interpret $\mathbf{f}$ as the limit of two force couples, and verify that their torques cancel.

Solution. (a) The linearised equations of motion obtained in the lecture are

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j}, \quad (10)$$

together with the boundary condition $A_{ijkl}(\partial u_k/\partial x_l)\hat{n}_j = \bar{S}_{ij}\hat{n}_j$ on ∂M . Substituting $\bar{S}_{ij} = M_{ij}(t)\delta(\mathbf{x})$, and noting that M_{ij} does not depend on position, the right hand side of eq. (10) becomes

$$f_i(\mathbf{x}, t) = -M_{ij}(t) \frac{\partial \delta}{\partial x_j}(\mathbf{x}), \quad (11)$$

while $\delta(\mathbf{x})$ vanishes identically on ∂M because the source lies in the interior, so the boundary condition reduces to the traction-free condition $A_{ijkl}(\partial u_k/\partial x_l)\hat{n}_j = 0$. The problem is therefore identical to that for the same elastic body driven by the body force $\mathbf{f}$, which is why the lecture refers to the right hand side of eq. (10) as an effective body force. The derivative of the delta function is understood in the sense of distributions: for any smooth ϕ ,

$$\int_M \phi \frac{\partial \delta}{\partial x_j} d^3\mathbf{x} = - \int_M \frac{\partial \phi}{\partial x_j} \delta d^3\mathbf{x} = - \frac{\partial \phi}{\partial x_j}(\mathbf{0}), \quad (12)$$

where the surface term arising from integration by parts vanishes because δ is supported only at the origin.

(b) Applying eq. (12) with $\phi = 1$ gives the net force

$$\int_M f_i d^3\mathbf{x} = -M_{ij} \int_M \frac{\partial \delta}{\partial x_j} d^3\mathbf{x} = M_{ij} \frac{\partial(1)}{\partial x_j} = 0, \quad (13)$$

and this holds for any M_{ij} , symmetric or not; it is a consequence of the source entering only as a divergence. For the torque about the origin we apply eq. (12) with $\phi = x_j$,

$$\int_M \epsilon_{ijk} x_j f_k d^3\mathbf{x} = -\epsilon_{ijk} M_{kl} \int_M x_j \frac{\partial \delta}{\partial x_l} d^3\mathbf{x} = \epsilon_{ijk} M_{kl} \left. \frac{\partial x_j}{\partial x_l} \right|_{\mathbf{x}=0} = \epsilon_{ijk} M_{kl} \delta_{jl} = \epsilon_{ijk} M_{kj}. \quad (14)$$

This is a contraction of the symmetric tensor M_{kj} with the alternating symbol, which is antisymmetric in j and k , and so it vanishes. Since the net force is zero, the torque about any other point is the same, and hence also zero. Conversely, if M_{ij} had an antisymmetric part, eq. (14) shows that the source would exert a net torque equal to minus twice the axial vector of that part. The symmetry of the stress glut, which in the lecture was a consequence of the invariance of the strain energy under superimposed rigid rotations, is thus exactly the condition that the source does not alter the angular momentum of the body, while the divergence form guarantees that it does not alter the linear momentum. An earthquake is an internal process, and cannot do either.

(c) For the double couple $M_{13} = M_{31} = M_0$ of Example 1, eq. (11) gives

$$f_1 = -M_0 \frac{\partial \delta}{\partial x_3}, \quad f_2 = 0, \quad f_3 = -M_0 \frac{\partial \delta}{\partial x_1}. \quad (15)$$

A derivative of the delta function is the limit of a difference of two shifted delta functions,

$$\frac{\partial \delta}{\partial x_3}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{1}{h} \left[\delta\left(\mathbf{x} + \frac{h}{2} \hat{\mathbf{e}}_3\right) - \delta\left(\mathbf{x} - \frac{h}{2} \hat{\mathbf{e}}_3\right) \right], \quad (16)$$

as follows by integrating both sides against a smooth ϕ and Taylor expanding. Hence

$$f_1 = \lim_{h \rightarrow 0} \frac{M_0}{h} \left[\delta\left(\mathbf{x} - \frac{h}{2} \hat{\mathbf{e}}_3\right) - \delta\left(\mathbf{x} + \frac{h}{2} \hat{\mathbf{e}}_3\right) \right], \quad (17)$$

which is a point force of magnitude $F = M_0/h$ directed along $+\hat{\mathbf{e}}_1$ at $\mathbf{x} = +\frac{h}{2}\hat{\mathbf{e}}_3$, together with an equal and opposite force at $\mathbf{x} = -\frac{h}{2}\hat{\mathbf{e}}_3$. This is a **force couple** with arm h and moment $Fh = M_0$, which remains finite as $h \rightarrow 0$; the forces act along the slip direction and are separated along the fault normal. The side $x_3 > 0$ is pushed in the $+\hat{\mathbf{e}}_1$ direction and the side $x_3 < 0$ in the $-\hat{\mathbf{e}}_1$ direction, consistent with the displacement jumping by $+D \hat{\mathbf{e}}_1$ on crossing the fault from $x_3 < 0$ to $x_3 > 0$. Its torque is

$$\frac{h}{2} \hat{\mathbf{e}}_3 \times F \hat{\mathbf{e}}_1 + \left(-\frac{h}{2} \hat{\mathbf{e}}_3\right) \times (-F \hat{\mathbf{e}}_1) = Fh \hat{\mathbf{e}}_3 \times \hat{\mathbf{e}}_1 = M_0 \hat{\mathbf{e}}_2. \quad (18)$$

In exactly the same way f_3 is a couple of forces $\pm F \hat{\mathbf{e}}_3$ at $\pm \frac{h}{2} \hat{\mathbf{e}}_1$, with forces along the fault normal separated along the slip direction, and torque

$$Fh \hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_3 = -M_0 \hat{\mathbf{e}}_2. \quad (19)$$

The two torques cancel, as they must by part (b). The first couple is the one we might have written down intuitively from the picture of slip on the fault; the second, which corresponds to slip on the auxiliary plane, is forced upon us by the requirement that the source exerts no net torque, and it is the pair together that constitutes the **double couple**. A single couple would set the body spinning.

Example 3: An explosive source generates only P-waves. An explosion at the origin of a homogeneous isotropic whole space, with Lamé parameters λ and μ and density ρ , is modelled by the isotropic stress glut

$$\bar{S}_{ij}(\mathbf{x}, t) = M_0 \delta_{ij} \delta(\mathbf{x}) H(t).$$

Find the equivalent body force, write down the resulting equation of motion for $\mathbf{u}$, and by taking its curl show that the displacement field is irrotational for all time. Deduce that only P-waves are generated.

Solution. The equivalent body force is

$$f_i = -\frac{\partial}{\partial x_j} [M_0 \delta_{ij} \delta(\mathbf{x}) H(t)] = -M_0 H(t) \frac{\partial \delta}{\partial x_i}(\mathbf{x}), \quad \text{i.e.} \quad \mathbf{f} = -M_0 H(t) \nabla \delta(\mathbf{x}), \quad (20)$$

which is the gradient of a scalar. For a homogeneous isotropic body we substitute eq. (1) into the elastic operator and use the constancy of λ and μ ,

$$\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = \lambda \frac{\partial^2 u_k}{\partial x_i \partial x_k} + \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j} + \mu \frac{\partial^2 u_j}{\partial x_i \partial x_i} = (\lambda + \mu) \frac{\partial}{\partial x_i} (\nabla \cdot \mathbf{u}) + \mu \nabla^2 u_i, \quad (21)$$

so that the equation of motion is the one given in Lecture 13 for a homogeneous isotropic body, now with a body force,

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} - (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) - \mu \nabla^2 \mathbf{u} = -M_0 H(t) \nabla \delta(\mathbf{x}), \quad (22)$$

to be solved with $\mathbf{u} = \partial \mathbf{u} / \partial t = \mathbf{0}$ at $t = 0$. Now take the curl of eq. (22) and write $\boldsymbol{\omega} = \nabla \times \mathbf{u}$. The curl of a gradient vanishes, which disposes of both the $(\lambda + \mu)$ term and the body force, while the curl commutes with ∇^2 and with $\partial^2 / \partial t^2$ because the coefficients are constant. We are left with

$$\rho \frac{\partial^2 \boldsymbol{\omega}}{\partial t^2} - \mu \nabla^2 \boldsymbol{\omega} = \mathbf{0}, \quad \boldsymbol{\omega}(\mathbf{x}, 0) = \mathbf{0}, \quad \frac{\partial \boldsymbol{\omega}}{\partial t}(\mathbf{x}, 0) = \mathbf{0}, \quad (23)$$

the initial conditions following by taking the curl of those for $\mathbf{u}$, which hold at every point of the body. Each Cartesian component of $\boldsymbol{\omega}$ thus satisfies the homogeneous scalar wave equation with speed $\beta = \sqrt{\mu / \rho}$ and vanishing initial data, and the only such solution is zero. To see this, consider the energy

$$\mathcal{E}(t) = \frac{1}{2} \int \left(\rho \left\| \frac{\partial \boldsymbol{\omega}}{\partial t} \right\|^2 + \mu \frac{\partial \omega_i}{\partial x_j} \frac{\partial \omega_i}{\partial x_j} \right) d^3 \mathbf{x}, \quad (24)$$

whose time derivative, after an integration by parts, is $\int \dot{\omega}_i (\rho \ddot{\omega}_i - \mu \nabla^2 \omega_i) d^3 \mathbf{x} = 0$; the surface term at infinity vanishes since, at any finite time, the disturbance has only travelled a finite distance from the source. Thus $\mathcal{E}(t) = \mathcal{E}(0) = 0$, and as $\mathcal{E}$ is a sum of non-negative terms both $\partial \boldsymbol{\omega} / \partial t$ and the gradient of $\boldsymbol{\omega}$ vanish; $\boldsymbol{\omega}$ is therefore constant in space and time, and equal to its initial value $\mathbf{0}$. The displacement is **irrotational** at all times.

In a whole space an irrotational field can be written as a gradient, $\mathbf{u} = \nabla\phi$. Substituting into eq. (22), and using $\nabla^2\nabla\phi = \nabla\nabla^2\phi$ and $\nabla \cdot \nabla\phi = \nabla^2\phi$, every term becomes a gradient,

$$\nabla \left[\rho \frac{\partial^2 \phi}{\partial t^2} - (\lambda + 2\mu) \nabla^2 \phi + M_0 H(t) \delta(\mathbf{x}) \right] = \mathbf{0}. \quad (25)$$

The bracket is therefore a function of t alone, which can be absorbed into ϕ without changing $\mathbf{u}$, and so

$$\rho \frac{\partial^2 \phi}{\partial t^2} - (\lambda + 2\mu) \nabla^2 \phi = -M_0 H(t) \delta(\mathbf{x}). \quad (26)$$

This is the scalar wave equation with wave speed $\alpha = \sqrt{(\lambda + 2\mu)/\rho}$, the P-wave speed of Lecture 15: the explosion radiates a spherically symmetric P-wave and nothing else. The isotropic moment tensor $M_0 \delta_{ij}$ is invariant under all rotations, so there is no fault plane and no preferred direction in the radiation. This should be contrasted with the double couple of Example 1, whose moment tensor is traceless. In practice explosions do produce S-waves, because the Earth is neither homogeneous nor unbounded and P-waves are partly converted to S-waves at the free surface and at internal discontinuities, but these are secondary. The size of the isotropic part of the moment tensor, essentially zero for an earthquake, is one of the discriminants used to distinguish nuclear tests from earthquakes.

Example 4: Finite rupture and the moment rate function. A rectangular fault occupies the region $0 \leq x_1 \leq L$, $0 \leq x_2 \leq W_f$ of the plane $x_3 = 0$ in a body with constant shear modulus μ . Rupture begins at $x_1 = 0$ at time $t = 0$ and propagates in the x_1 -direction at constant speed v_r , with the slip at each point taking place instantaneously as the rupture front passes,

$$w_1(x_1, x_2, t) = D H\left(t - \frac{x_1}{v_r}\right), \quad w_2 = w_3 = 0.$$

Find the time-dependent scalar moment $M_0(t)$ and the **moment rate** $\dot{M}_0(t)$, and show that the latter is a boxcar of duration L/v_r . Estimate the rupture duration and the total moment for $L = 100$ km, $v_r = 3$ km s⁻¹, $W_f = 20$ km, $D = 2$ m and $\mu = 3 \times 10^{10}$ Pa.

Solution. From eq. (4) the stress glut has the single independent component $\bar{S}_{13} = \bar{S}_{31} = \mu w_1 \delta(x_3)$, and so by the definition of the moment tensor in Example 1 the only non-zero components are

$$M_{13}(t) = M_{31}(t) = \mu \int_{\Sigma} w_1 dS \equiv M_0(t), \quad (27)$$

which defines the time-dependent scalar moment. The moment tensor has the double-couple form of Example 1 at every instant, with the whole time dependence carried by $M_0(t)$. Substituting the assumed slip, and performing the trivial integral over x_2 ,

$$M_0(t) = \mu W_f D \int_0^L H\left(t - \frac{x_1}{v_r}\right) dx_1. \quad (28)$$

The step function equals one where $x_1 < v_r t$ and zero elsewhere, so the integral is the length of the ruptured portion of the fault at time t , namely

$$M_0(t) = \begin{cases} 0, & t < 0, \\ \mu W_f D v_r t, & 0 \leq t \leq T_r, \\ \mu W_f L D, & t > T_r, \end{cases} \quad T_r = \frac{L}{v_r}. \quad (29)$$

The moment grows linearly while the rupture front crosses the fault and then stays at its final value $\mu \mathcal{A} D$, with $\mathcal{A} = L W_f$, in agreement with Example 1. Differentiating under the integral sign, and using $H' = \delta$,

$$\dot{M}_0(t) = \mu W_f \int_0^L \frac{\partial w_1}{\partial t} dx_1 = \mu W_f D \int_0^L \delta\left(t - \frac{x_1}{v_r}\right) dx_1 = \mu W_f D v_r \int_0^{T_r} \delta(t - \tau) d\tau, \quad (30)$$

where we substituted $\tau = x_1/v_r$. The final integral is one if $0 < t < T_r$ and zero otherwise, and so

$$\dot{M}_0(t) = \mu W_f D v_r [H(t) - H(t - T_r)], \quad (31)$$

a boxcar of height $\mu W_f D v_r$ and duration T_r , in agreement with the derivative of eq. (29). Its area is $\mu W_f D v_r T_r = \mu \mathcal{A} D$, the total moment, as it must be.

For the given values the rupture duration is

$$T_r = \frac{100 \text{ km}}{3 \text{ km s}^{-1}} \approx 33 \text{ s}, \quad (32)$$

and the total moment is

$$M_0 = \mu \mathcal{A} D = (3 \times 10^{10} \text{ Pa}) (10^5 \text{ m}) (2 \times 10^4 \text{ m}) (2 \text{ m}) = 1.2 \times 10^{20} \text{ N m}, \quad (33)$$

with a moment rate of $3.6 \times 10^{18} \text{ N m s}^{-1}$ during the rupture. For orientation, the moment magnitude is defined by $M_w = \frac{2}{3} [\log_{10}(M_0/\text{N m}) - 9.1]$, which here gives $M_w \approx 7.3$, a large earthquake and a plausible size for a rupture of these dimensions. The moment rate function, often called the source time function, is what a distant seismometer effectively records: the far-field displacement radiated by a point source is proportional to $\dot{M}_0$, so the pulse from this event would be roughly 33 s long. Two refinements are easily incorporated. If the slip at each point rises over a finite rise time τ_r rather than instantaneously, $\dot{M}_0$ becomes the convolution of the boxcar with a second boxcar of width τ_r , i.e. a trapezoid of total duration $T_r + \tau_r$. And for an observer in the direction of rupture propagation the pulse is compressed, as successive parts of the fault radiate from progressively closer points, an effect known as directivity. Note finally that the point-source approximation of Example 1 requires wavelengths large compared with L , which for this event means periods well in excess of T_r .

Example 5: Well- and ill-posedness of polynomial interpolation. Given distinct points $x_1, \dots, x_n$ and values $y_1, \dots, y_n$, we seek a polynomial $p(x) = \sum_{k=0}^m c_k x^k$ with $p(x_i) = y_i$. Write the problem as a linear system $\mathbf{V}\mathbf{c} = \mathbf{y}$ for the coefficient vector $\mathbf{c}$, and then:

- (a) for $n = 3$ and $m = 2$, show explicitly that a solution exists, is unique, and depends continuously on $\mathbf{y}$;
- (b) for $n = 3$ and $m = 1$, derive the condition that the data must satisfy for a solution to exist;
- (c) for $n = 2$ and $m = 2$, exhibit the one-parameter family of solutions;
- (d) for equally spaced points on $[0, 1]$ with $m = n - 1$, compute the condition number of $\mathbf{V}$ for $n = 5, 10, 15$ and 20 , and comment on what continuous dependence is worth in practice.

Solution. The interpolation conditions are linear in the unknown coefficients,

$$\sum_{k=0}^m x_i^k c_k = y_i, \quad i = 1, \dots, n, \quad \text{i.e.} \quad \mathbf{V}\mathbf{c} = \mathbf{y}, \quad V_{ik} = x_i^k, \quad (34)$$

where $\mathbf{V}$ is the $n \times (m + 1)$ **Vandermonde matrix**. In the language of the lecture the inputs are $\mathbf{y}$ and the outputs are $\mathbf{c}$. A solution exists if and only if $\mathbf{y}$ lies in the column space of $\mathbf{V}$, it is unique if and only if the null space of $\mathbf{V}$ is trivial, and, whenever the solution is unique, it depends continuously on $\mathbf{y}$ because it is then given by a linear map between finite-dimensional spaces, and every such map is continuous.

(a) With $n = 3$ and $m = 2$ the matrix is square, and subtracting the first row from the other two,

$$\det \mathbf{V} = \det \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} = \det \begin{pmatrix} x_2 - x_1 & x_2^2 - x_1^2 \\ x_3 - x_1 & x_3^2 - x_1^2 \end{pmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2), \quad (35)$$

where the last step factors $x_j^2 - x_1^2 = (x_j - x_1)(x_j + x_1)$ from each row. This is non-zero for distinct points, so $\mathbf{V}$ is invertible and the problem has the unique solution $\mathbf{c} = \mathbf{V}^{-1}\mathbf{y}$ for every $\mathbf{y}$. The solution can also be written down directly in Lagrange form,

$$p(x) = \sum_{i=1}^3 y_i \ell_i(x), \quad \ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}, \quad (36)$$

since $\ell_i(x_j) = \delta_{ij}$; this establishes existence, and uniqueness follows because the difference of two solutions is a quadratic with three distinct roots, hence identically zero. Continuous dependence is quantified by

$$\|\delta \mathbf{c}\| \leq \|\mathbf{V}^{-1}\| \|\delta \mathbf{y}\|, \quad \frac{\|\delta \mathbf{c}\|}{\|\mathbf{c}\|} \leq \text{cond}(\mathbf{V}) \frac{\|\delta \mathbf{y}\|}{\|\mathbf{y}\|}, \quad \text{cond}(\mathbf{V}) = \|\mathbf{V}\| \|\mathbf{V}^{-1}\|, \quad (37)$$

where $\|\cdot\|$ is the matrix norm induced by the Euclidean vector norm and $\text{cond}(\mathbf{V})$ is the **condition number**, which for this norm is the ratio of the largest to the smallest

singular value of $\mathbf{V}$. The second inequality follows from the first together with $\|\mathbf{y}\| \leq \|\mathbf{V}\| \|\mathbf{c}\|$. The problem is well-posed.

(b) With $n = 3$ and $m = 1$ we have three equations $c_0 + c_1 x_i = y_i$ for two unknowns. The two columns of $\mathbf{V}$, namely $(1, 1, 1)^T$ and $(x_1, x_2, x_3)^T$, are linearly independent, so the column space is a plane in $\mathbb{R}^3$ and $\mathbf{y}$ must lie in it. The plane is the orthogonal complement of any non-zero vector $\mathbf{a}$ with $\mathbf{a}^T \mathbf{V} = \mathbf{0}$, and

$$\mathbf{a} = (x_2 - x_3, x_3 - x_1, x_1 - x_2)^T \quad (38)$$

does the job, since its components sum to zero and $(x_2 - x_3)x_1 + (x_3 - x_1)x_2 + (x_1 - x_2)x_3 = 0$. Existence therefore requires

$$(x_2 - x_3)y_1 + (x_3 - x_1)y_2 + (x_1 - x_2)y_3 = 0, \quad (39)$$

and this condition is also sufficient. Rearranging, it reads $(y_3 - y_1)(x_2 - x_1) = (y_2 - y_1)(x_3 - x_1)$, which states that the slope from the first point to the third equals the slope from the first to the second: the three points (x_i, y_i) must be collinear. Equivalently, the determinant of the matrix with rows $(1, x_i, y_i)$ vanishes. For equally spaced points $x_i = x_1 + (i - 1)h$ the condition is $y_1 - 2y_2 + y_3 = 0$, the vanishing of the second difference. When it holds, the solution is $c_1 = (y_2 - y_1)/(x_2 - x_1)$ and $c_0 = y_1 - c_1 x_1$. Data chosen at random satisfy eq. (39) with probability zero, so for noisy measurements the problem as posed has no solution, and one is led, as noted in the lecture, to relax what is meant by a solution, for instance to the $\mathbf{c}$ that minimises $\|\mathbf{V}\mathbf{c} - \mathbf{y}\|$.

(c) With $n = 2$ and $m = 2$ there are two equations for three unknowns. A vector $\mathbf{c}$ lies in the null space if and only if the corresponding quadratic vanishes at x_1 and x_2 , that is, if and only if $p(x) = \alpha(x - x_1)(x - x_2)$ for some α , so the null space is spanned by

$$\mathbf{c}_0 = (x_1 x_2, -(x_1 + x_2), 1)^T, \quad (40)$$

as may be checked directly: $x_1 x_2 - (x_1 + x_2)x_i + x_i^2 = (x_i - x_1)(x_i - x_2) = 0$. A particular solution is the straight line through the two points, and the general solution is

$$p(x) = y_1 + \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) + \alpha(x - x_1)(x - x_2), \quad (41)$$

or in terms of coefficients, with $s = (y_2 - y_1)/(x_2 - x_1)$,

$$c_2 = \alpha, \quad c_1 = s - \alpha(x_1 + x_2), \quad c_0 = y_1 - s x_1 + \alpha x_1 x_2, \quad (42)$$

for arbitrary α . Every quadratic through the two data points fits them equally well, and the data provide no information whatever about α . This is the underdetermined case of the lecture, and it is the situation in which almost all geophysical inverse problems find themselves. To select one member of the family we must add information not contained in the data, for example by choosing the α that minimises $\|\mathbf{c}\|$, and later lectures will have much to say about how such choices are made and justified.

(d) For $m = n - 1$ the matrix is square and, by the argument of part (a) generalised to n points, invertible, so the problem is well-posed for every n . Taking equally spaced points $x_i = (i - 1)/(n - 1)$ on $[0, 1]$ and computing the condition number in

the Euclidean norm numerically,¹ we find

n	5	10	15	20	
$\text{cond}(\mathbf{V})$	6.9×10^2	1.5×10^7	4.0×10^{11}	1.1×10^{16}	(43)

The growth is exponential in n . By eq. (37) the condition number bounds the factor by which relative errors in the data can be amplified in the coefficients, and the bound is attained for suitably chosen perturbations. For $n = 10$, data accurate to one part in 10^7 can produce coefficients with errors of order one hundred per cent; for $n = 20$, data accurate to sixteen significant figures, i.e. to the full precision of double arithmetic, can produce coefficients with no correct digits at all. The problem is well-posed in the strict sense of the lecture, since the solution does depend continuously on the data, but the constant in that continuity is so large that the solution is practically useless. Once measurement errors and finite precision are admitted, the distinction between an ill-posed problem and a badly conditioned one all but disappears, and both call for the same kind of remedy. It is worth adding that the ill-conditioning here lies in the choice of unknowns: the monomials x^k on $[0, 1]$ become nearly parallel as k grows, so the columns of $\mathbf{V}$ are nearly linearly dependent. The values of the interpolating polynomial are far better determined than its monomial coefficients, and a wiser parameterisation, for instance in terms of Chebyshev polynomials, greatly reduces the condition number. Choosing the model parameters well is often the first step in taming an inverse problem.

¹ For $n = 20$ the condition number is comparable with the reciprocal of double-precision machine epsilon, and a floating-point computation of it is itself uncertain at the level of a few per cent; a routine such as `numpy.linalg.cond` returns 1.2×10^{16} . The value quoted was obtained by inverting $\mathbf{V}$ in exact rational arithmetic before taking norms.